

HDOC 4

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Question

The following problem is to find the product of two integers without using the multiplication operator *. Your teacher has given an incorrect algorithm and an incorrect flowchart for this problem (Refer Slideshare). Study both carefully and complete the tasks given below. *[Hint: Use repeated addition]*

Requirements: The logic must use the minimum possible number of additions and should correctly handle:

- positive x positive
- positive x negative
- negative x positive
- negative x negative
- multiplication by zero

Tasks:

- Identify the mistakes in the given algorithm.
- Rewrite the corrected algorithm.
- Identify the mistakes in the given flowchart.
- Redraw the corrected flowchart.
- Verify your corrected algorithm and flow chart using the given test cases.

Test Case	Input	Expected Output	Minimum Additions
1	12 3	Product: 36	3
2	3 12	Product: 36	3
3	-6 5	Product: -30	5
4	6 -5	Product: -30	5
5	-9 -4	Product: 36	4
6	-4 -9	Product: 36	4
7	0 8	Product: 0	0
8	8 0	Product: 0	0

Test Case	Input	Expected Output	Minimum Additions
9	0 -7	Product: 0	0
10	-7 0	Product: 0	0
11	0 0	Product: 0	0
12	1 -7	Product: -7	1
13	-1 -9	Product: 9	1

Correct Logic / Algorithm

1. Read two integers A and B.
2. If A = 0 or B = 0, set product = 0 and additions = 0.
3. Otherwise, take the absolute values of A and B.
4. Choose the smaller absolute value as the repetition count. This guarantees the minimum number of additions.
5. Choose the larger absolute value as the value to be added repeatedly.
6. Start product = 0 and add the larger value exactly 'count' times.
7. If A and B have opposite signs, make the product negative; otherwise keep it positive.
8. Display the product and the number of additions.

C Program

```
#include <stdio.h>

int main() {
    int a, b, x, y;
    int addValue, count;
    int product = 0, additions = 0;
    int i;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    if (a == 0 || b == 0) {
        printf("Product: 0\n");
        printf("Minimum additions: 0\n");
        return 0;
    }

    x = (a < 0) ? -a : a;
    y = (b < 0) ? -b : b;

    if (x < y) {
        count = x;
        addValue = y;
    } else {
        count = y;
        addValue = x;
    }

    for (i = 0; i < count; i++) {
        product = product + addValue;
        additions++;
    }

    if ((a < 0 && b > 0) || (a > 0 && b < 0))
        product = -product;

    printf("Product: %d\n", product);
    printf("Minimum additions: %d\n", additions);

    return 0;
}
```

Output / Verification

The program gives the required product and uses the minimum repeated additions for every supplied test case.

Input	Program Output	Additions
12 3	Product: 36	3
3 12	Product: 36	3
-6 5	Product: -30	5
6 -5	Product: -30	5
-9 -4	Product: 36	4
-4 -9	Product: 36	4
0 8	Product: 0	0
8 0	Product: 0	0
0 -7	Product: 0	0
-7 0	Product: 0	0
0 0	Product: 0	0
1 -7	Product: -7	1
-1 -9	Product: 9	1

Example Console Output

```
Enter two integers: 12 3
Product: 36
Minimum additions: 3

Enter two integers: 6 -5
Product: -30
Minimum additions: 5

Enter two integers: -9 -4
Product: 36
Minimum additions: 4

Enter two integers: 0 8
Product: 0
Minimum additions: 0
```